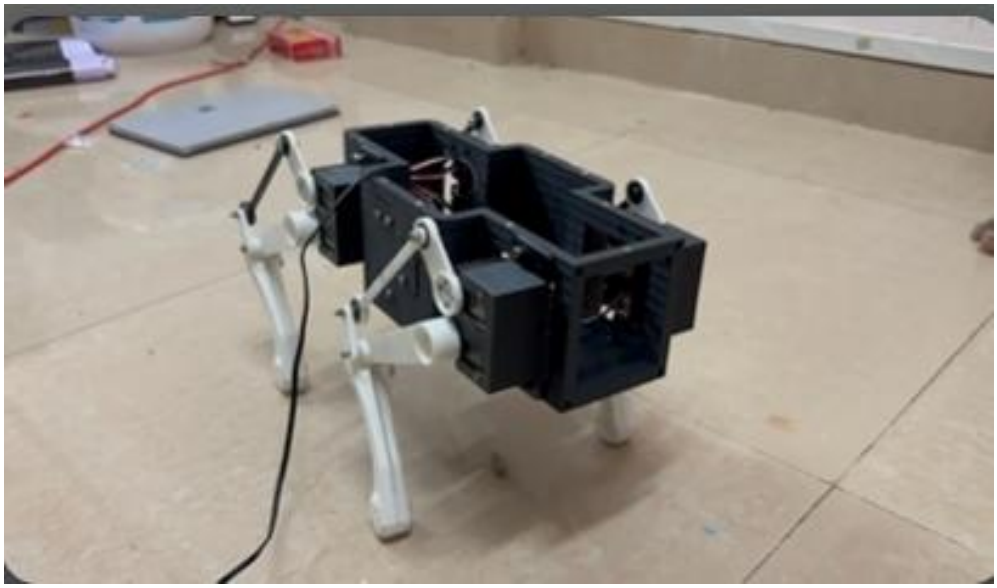
QUADRUPED ROBOT WITH 5-BAR MECHANISMS

In this project a smaller quadruped robot with just 8 degrees of freedom and was designed and fabricated from scratch and we were able to implement the basic forward and inverse kinematics for sitting and standing. The quadruped was actuated with the help of ST3215 Servo motors which were driven by a Waveshare serial servo motor driver. All the parts were 3D printed using PLA filament and were assembled together using mechanical fasteners and custom soldering techniques.



The link lengths were designed using CAD and based on their lengths, the forward and inverse kinematic equations were derived for a 5 bar mechanism and implemented in a MATLAB code.

Servo control was achieved using ESP32 and Raspberry Pi.